



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

TOWER OF HANOI

Recursive Problem Solving Using Divide and Conquer

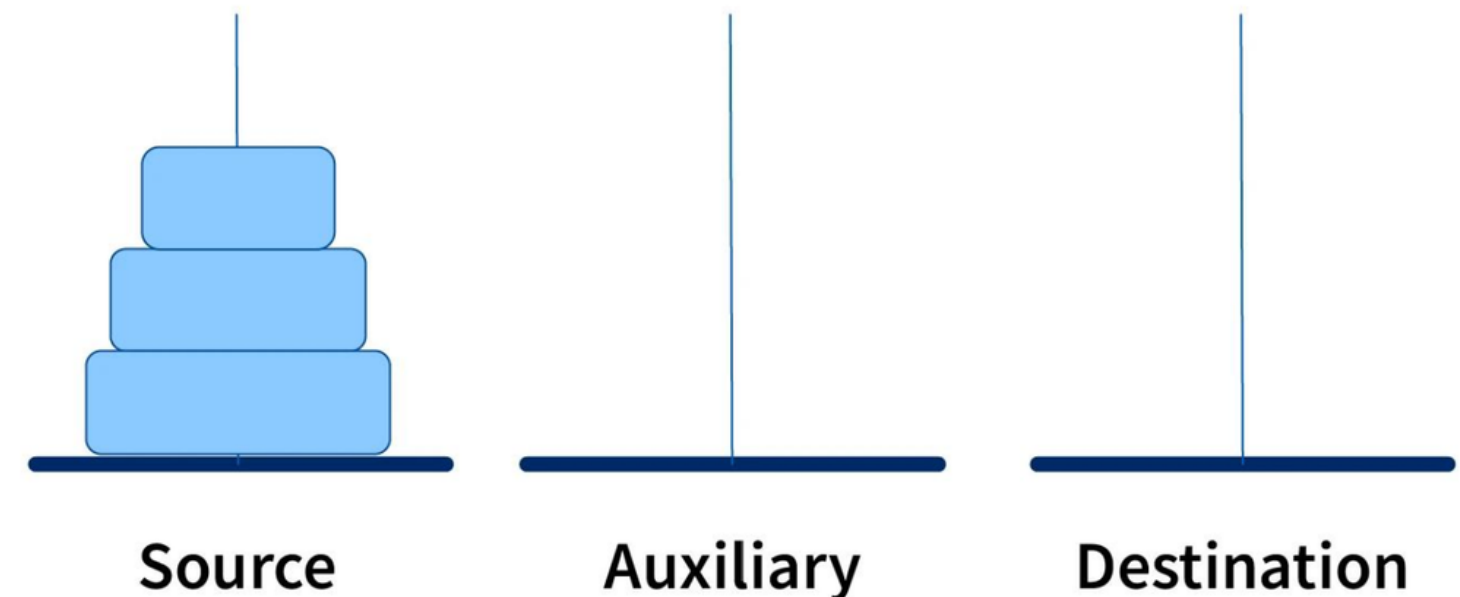
Submitted by:
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INTRODUCTION

1. TOWER OF HANOI IS A FAMOUS MATHEMATICAL PUZZLE.
2. IT WAS INTRODUCED BY FRENCH MATHEMATICIAN ÉDOUARD LUCAS IN 1883.
3. IT CONSISTS OF THREE RODS AND N DISKS OF DIFFERENT SIZES.
4. THE OBJECTIVE IS TO MOVE ALL DISKS FROM THE SOURCE ROD TO THE DESTINATION ROD.

COMPONENTS OF TOWER OF HANOI

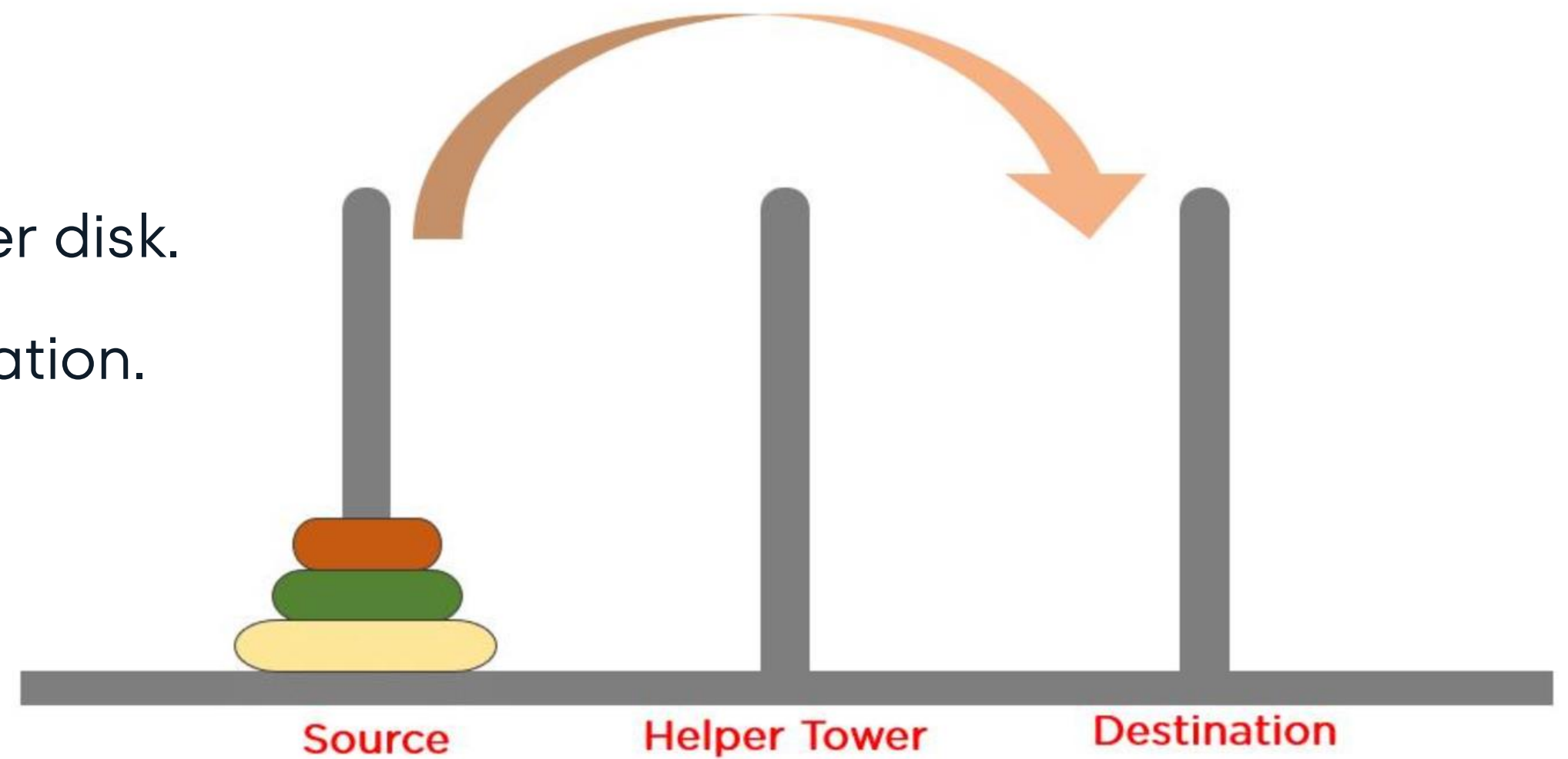
- **Source Rod (A):** Initially contains all disks.
- **Auxiliary Rod (B):** Used as temporary storage.
- **Destination Rod (C):** Final position of all disks.



RULES

1. Move only one disk at a time.
2. Only the top disk can be moved.
3. Never place a larger disk on a smaller disk.
4. Move all disks from Source to Destination.

The Objective of this puzzle, is to move all the rings from source tower to the destination tower.



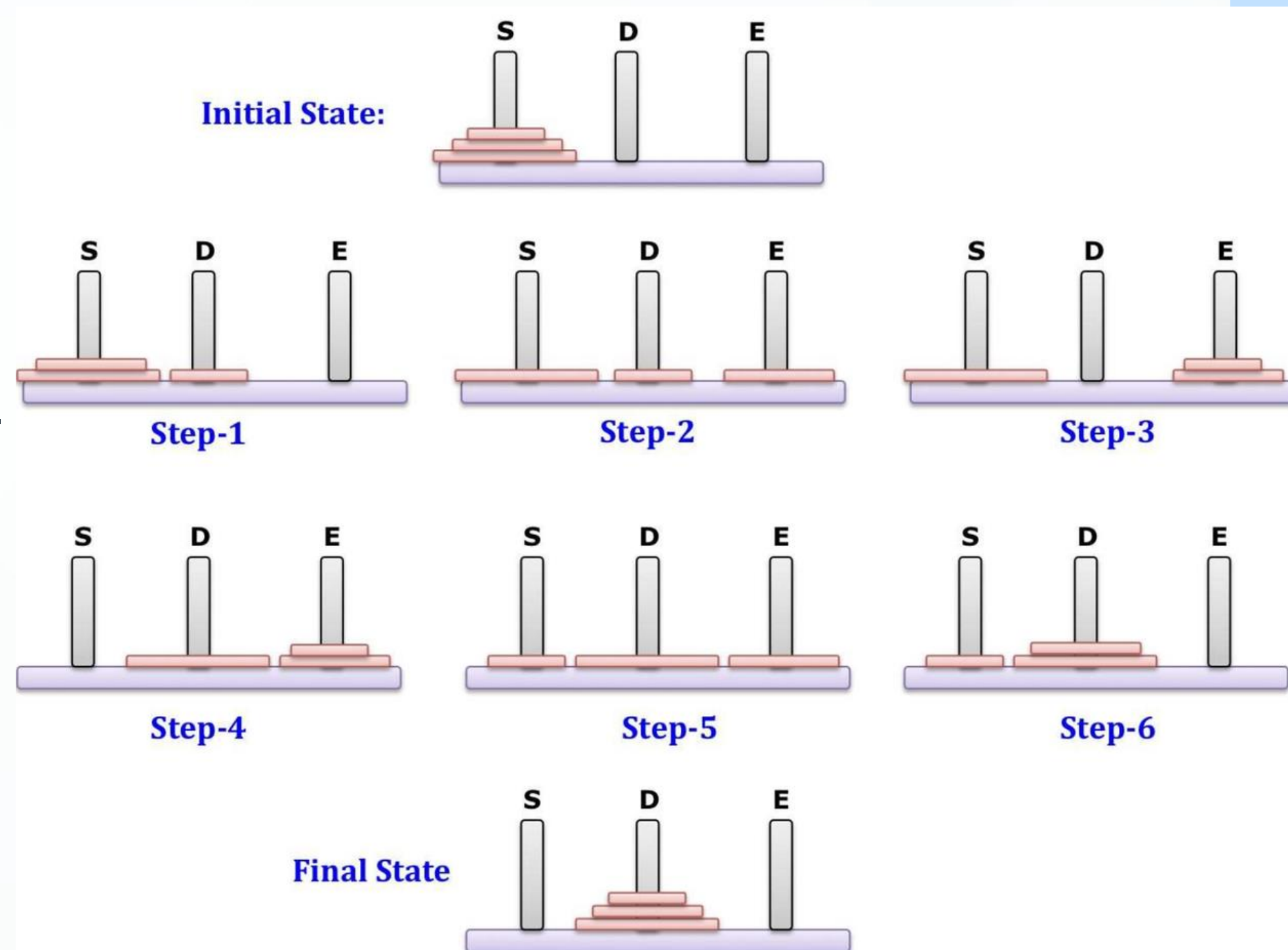
ALGORITHM

If $n = 1$

- Move disk from Source $\rightarrow$ Destination.

Otherwise:

1. Move $n-1$ disks from Source $\rightarrow$ Auxiliary.
2. Move largest disk to Destination.
3. Move $n-1$ disks from Auxiliary $\rightarrow$ Destination.



WORKING PROCESS (3 DISKS)

Initial

A: 3 2 1

B:

C:

Step 1

- Move Disk 1 → C

Step 2

- Move Disk 2 → B

Step 3

- Move Disk 1 → B

Step 4

- Move Disk 3 → C

Step 5

- Move Disk 1 → A

Step 6

- Move Disk 2 → C

Step 7

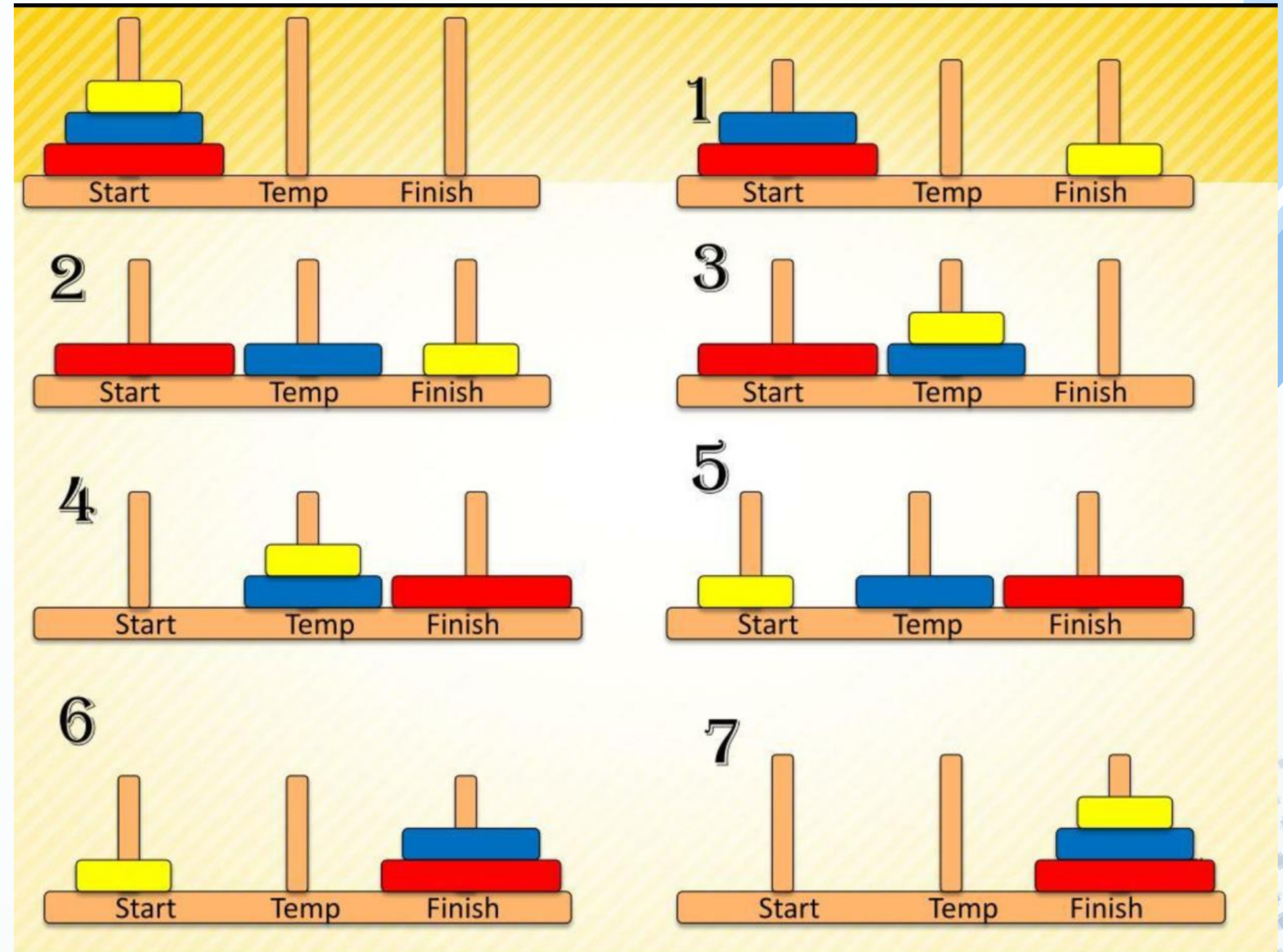
- Move Disk 1 → C

Final

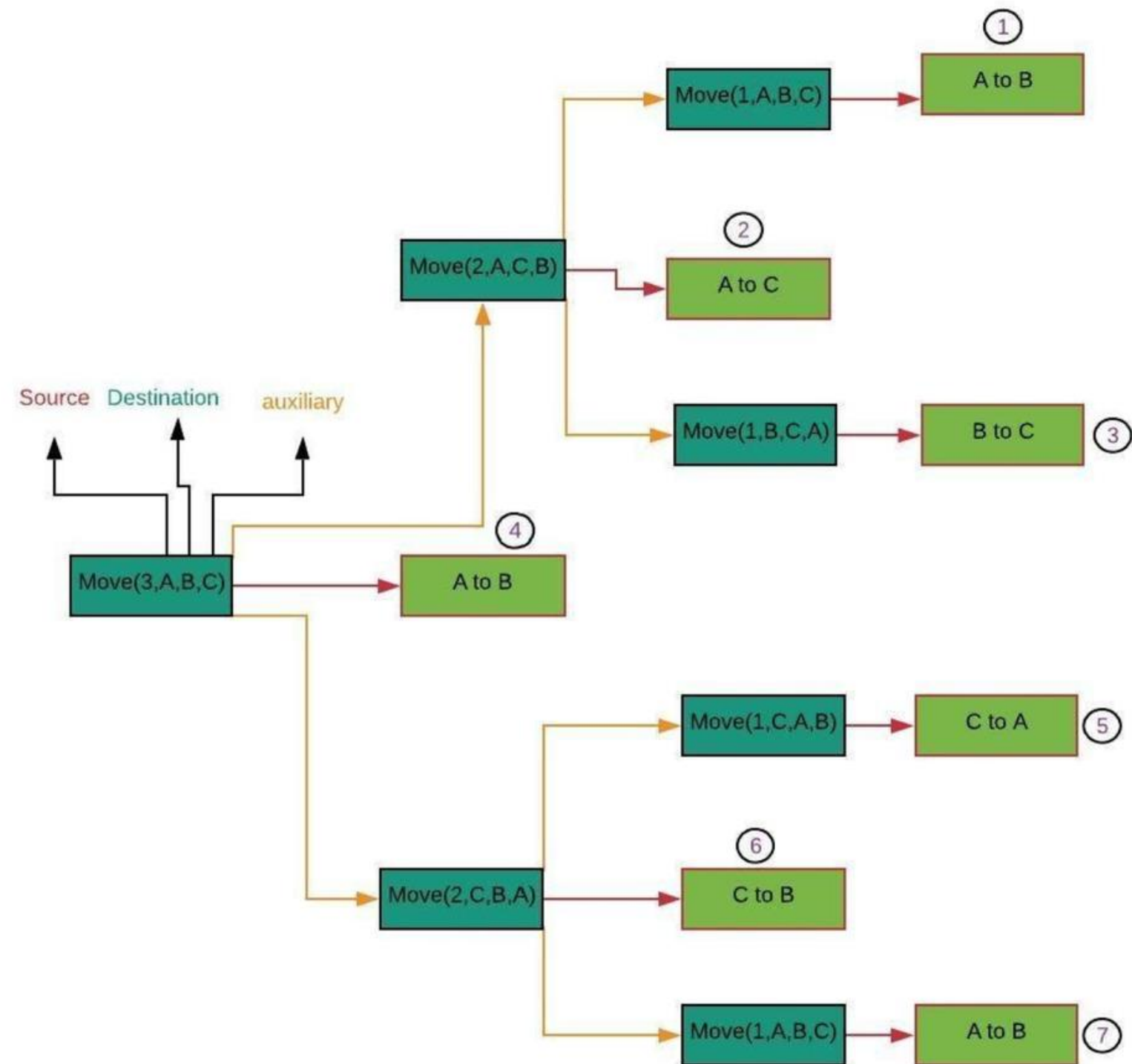
A:

B:

C: 3 2 1



RECURSION TREE





C PROGRAM

```
#include<stdio.h>
```

```
void tower(int n,char source,char aux,char dest)
{
    if(n==1)
    {
        printf("Move disk 1 from %c to %c\n",source,dest);
        return;
    }

    tower(n-1,source,dest,aux);

    printf("Move disk %d from %c to %c\n",n,source,dest);

    tower(n-1,aux,source,dest);
}

int main()
{
    tower(3,'A','B','C');
    return 0;
}
```


APPLICATIONS

- Teaching recursion
- Divide and Conquer algorithms
- Artificial Intelligence
- Puzzle solving
- Computer Science education
- Logical thinking development

CONCLUSION

- Tower of Hanoi is one of the most popular recursion problems.
- It demonstrates how a complex problem can be solved by breaking it into smaller subproblems.
- The minimum number of moves required is:
- **$2^n - 1$**
- It is an excellent example for learning recursion, algorithms, and problem-solving techniques.



THANK YOU